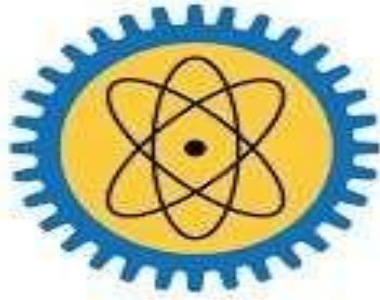




**BUSITEMA
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FACULTY OF ENGINEERING AND TECHNOLOGY

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SOLUTION OF $x^3 - 5x + 3 = 0$

*Secant Method, Newton–Raphson Method and Runge–Kutta Method
Including FOR and WHILE loop pseudocode and flowcharts*

Task;

Get the math questions solved in secant, runge-kutta, and newton Raphson

Get solutions for 'for' and 'while' loop for each of the equations. Draw flowchart and get pseudo code

1. Problem statement

Solve the nonlinear equation

$$f(x) = x^3 - 5x + 3 = 0$$

The equation has three real roots.

The following starting values are used: $x_0 = 0$, $x_1 = 1$ for the Secant method; $x_0 = 1$ for Newton–Raphson; and $x_0 = 1$ with $h = 0.1$ for the Runge–Kutta IV method after rewriting the root problem as the initial-value problem $y' = f(y)$, $y(0) = 1$.

2. Secant Method

Formula:

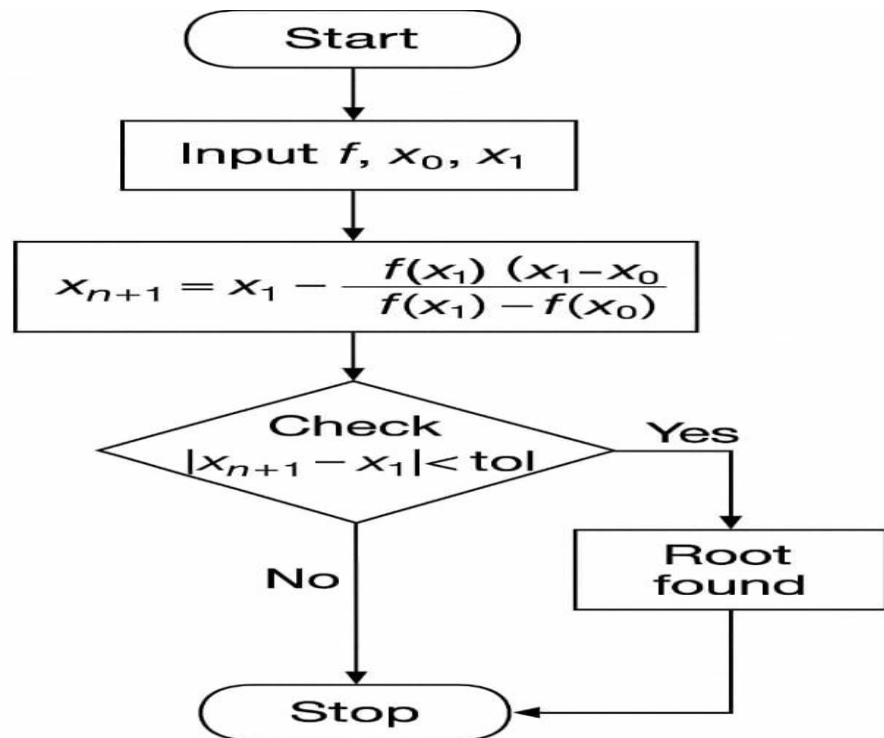
$$x_{n+1} = x_n - f(x_n) (x_n - x_{n-1}) / [f(x_n) - f(x_{n-1})]$$

Using $x_0 = 0$ and $x_1 = 1$ gives convergence to the root $x \approx 0.656620431$.

Iteration table:

N	x_{n-1}	x_n	x_{n+1}	$f(x_{n+1})$	$ \Delta x $
1	0	1	0.750000000	- 0.328125000	0.250000000
2	1	0.750000000	0.627906977	0.108028224	0.122093023
3	0.750000000	0.627906977	0.658147479	- 0.005655480	0.030240502
4	0.627906977	0.658147479	0.656643089	- 0.000083983	0.001504389
5	0.658147479	0.656643089	0.656620413	0.000000068	0.000022677
6	0.656643089	0.656620413	0.656620431	- 0.000000000	0.000000018

Result: $x \approx 0.656620431$



2.1. Secant Method — FOR loop pseudocode

START

$x_0 = 0, x_1 = 1;$

tolerance = 1.0000e-06

for n = 1: 50

$f_0 = x_0^3 - 5x_0 + 3$

$f_1 = x_1^3 - 5x_1 + 3$

$x_2 = x_1 - f_1(x_1 - x_0)/(f_1 - f_0)$

if $\text{abs}(x_2 - x_1) < \text{tolerance}$

break

end

$x_0 = x_1$

$x_1 = x_2$

end

fprintf ('Root = %.9f\n', x2);

END

2.2. Secant Method — WHILE loop pseudocode

START

$x_0 = 0;$

$x_1 = 1;$

tolerance = 1.0000e-06;

n = 0

while n < 50

$f_0 = x_0^3 - 5x_0 + 3$

$f_1 = x_1^3 - 5x_1 + 3$

```

x2 = x1 - f1*(x1-x0)/(f1-f0)

n = n + 1

if abs(x2-x1) < tolerance

    break

end

x0 = x1

x1 = x2

end

fprintf ('Root = %.9f\n', x2);
END

```

3. Newton–Raphson Method

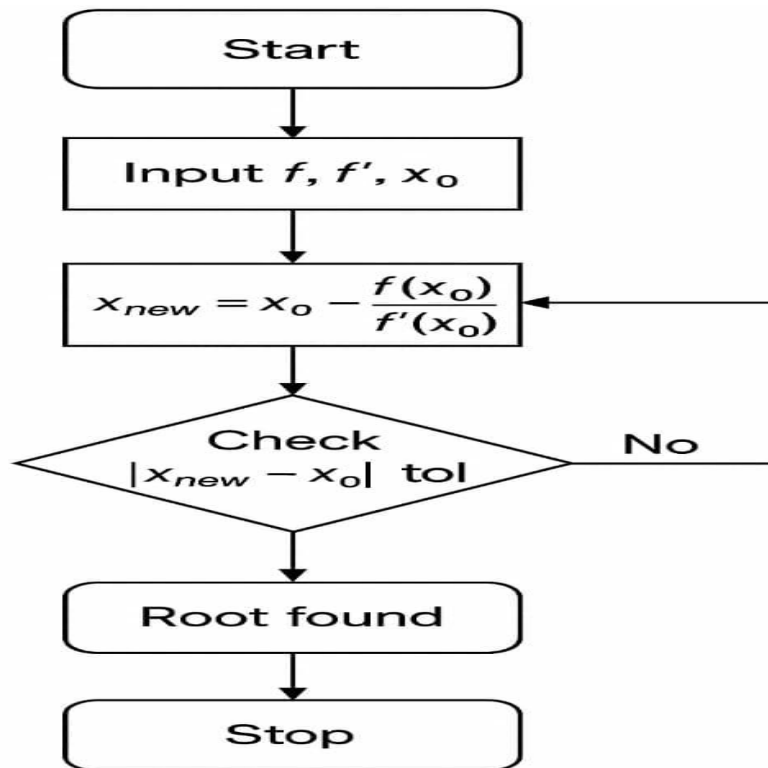
$$f'(x) = 3x^2 - 5$$

$$\text{Formula: } x_{n+1} = x_n - f(x_n)/f'(x_n)$$

N	x_n	x_{n+1}	$f(x_{n+1})$	$ \Delta x $
0	1	0.500000000	0.625000000	0.500000000
1	0.500000000	0.647058824	0.035619784	0.147058824
2	0.647058824	0.656572795	0.000176568	0.009513972
3	0.656572795	0.656620430	0.000000004	0.000047634
4	0.656620430	0.656620431	0.000000000	0.000000001

Result: $x \approx 0.656620431$

Newton's Raphson Method



3.1. Newton–Raphson — FOR loop pseudocode

start

$x = 1;$

tolerance = 1.0000e-06

for $n = 1: 50$

$fx = x^3 - 5x + 3$

$dfx = 3x^2 - 5$

$x_{new} = x - fx/dfx$

if $abs(x_{new}-x) < tolerance$

break

end

$x = x_{new}$

end

fprintf ('Root = %.9f\n', x_{new})

stop

3.2. Newton–Raphson — WHILE loop pseudocode

START

$x = 1;$

$\text{tolerance} = 1.0000\text{e-}06;$

$n = 0;$

while $n < 50$

$fx = x^3 - 5x + 3$

$dfx = 3x^2 - 5$

$x_{\text{new}} = x - fx/dfx$

$n = n + 1$

if $\text{abs}(x_{\text{new}} - x) < \text{tolerance}$

break

end

$x = x_{\text{new}}$

end

`fprintf ('Root = %.9f\n', xnew);`

END

4. Runge–Kutta 4th Order (RK4)

RK4 is an ODE solver rather than a conventional root-finding method. For the requested equation, we can use the auxiliary IVP $y' = y^3 - 5y + 3$, $y(0) = 1$ and demonstrate the RK4 algorithm. This does not by itself define a root of the original equation; the equilibrium values of the IVP satisfy $y^3 - 5y + 3 = 0$.

For $y' = F(y)$, the RK4 steps are:

$k_1 = F(y_n)$

$k_2 = F(y_n + h k_1/2)$

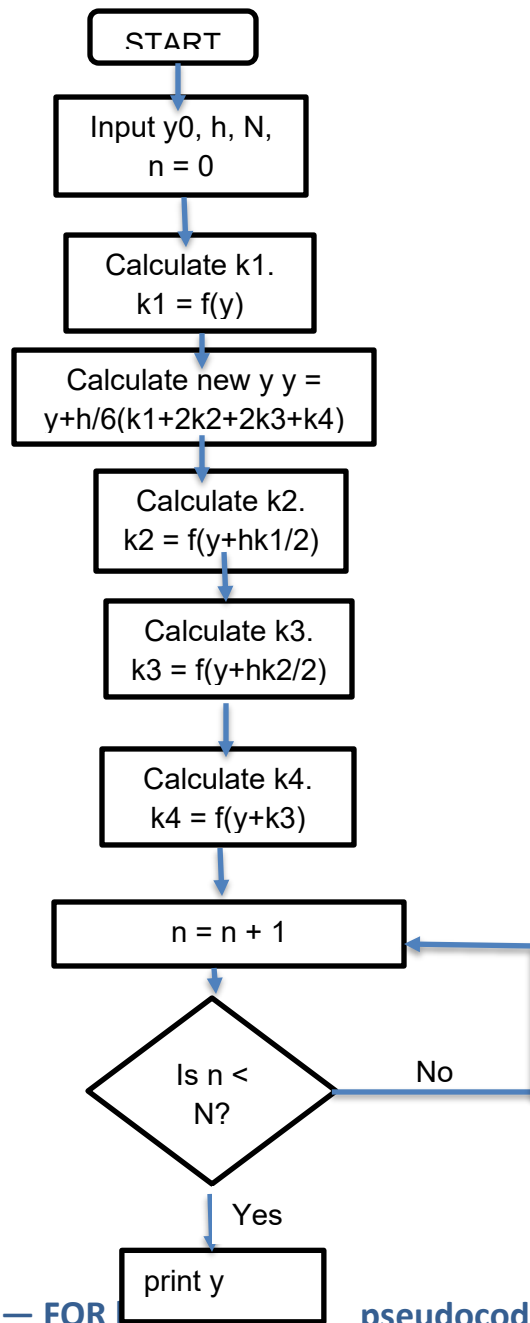
$k_3 = F(y_n + h k_2/2)$

$k_4 = F(y_n + h k_3)$

$y_{n+1} = y_n + (h/6) (k_1 + 2k_2 + 2k_3 + k_4)$

Use $h = 0.1$ and $y_0 = 1$. The RK4 iteration can be repeated with a FOR or WHILE loop.

3. Flow chart



4.1. RK4 — FOR LOOP pseudocode

START

y = 1, h = 0.1, N = 10

for n = 1 : N

$$k1 = y^3 - 5*y + 3$$

$$k2 = (y + h*k1/2)^3 - 5*(y + h*k1/2) + 3$$

END

```

k3 = (y + h*k2/2) ^3 - 5*(y + h*k2/2) + 3
k4 = (y + h*k3) ^3 - 5*(y + h*k3) + 3
y = y + (h/6) *(k1 + 2*k2 + 2*k3 + k4)
fprintf('n = %d, y = %.6f\n', n, y);
END

```

4.2. RK4 — WHILE loop pseudocode

```

START
y = 1;

h = 0.1;

n = 0;

N = 10

while n < N

    k1 = y^3 - 5*y + 3

    k2 = (y + h*k1/2) ^3 - 5*(y + h*k1/2) + 3

    k3 = (y + h*k2/2) ^3 - 5*(y + h*k2/2) + 3

    k4 = (y + h*k3) ^3 - 5*(y + h*k3) + 3

    y = y + (h/6) *(k1 + 2*k2 + 2*k3 + k4)

    n = n + 1

    fprintf('n = %d, y = %.6f\n', n, y);

end
END

```

Summary of results

For $x^3 - 5x + 3 = 0$, the three real roots are approximately:

- -2.490864
- 0.656620
- 1.834243